

# RAI Risk Taxonomy 2.0:

## Top-down evidence selection and staged classification of 1,725 global AI risks

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### Abstract

We present a reproducible, top-down account of the evidence and classification pipeline used to construct RAI Risk Taxonomy 2.0. The integrated evidence space contains 3,906,767 AI records: 1,749,159 papers, 125,372 policy reports, and 2,032,236 patents. A curated AI-risk evidence view contains 82,971 records (2.1% of the AI evidence space). Canonicalization, source crosswalking, and risk-card construction yields 1,726 candidate rows. A conservative exact-match check merges one duplicate, yielding 1,725 L4 cards. A staged hierarchy-blind classification procedure assigns all 1,725 cards (100.0%) to 50 L3 nodes. Two independent expert reviews yield one consensus label amendment and leave 83 cards (4.8%) marked **HOLD**. We specify the filter at each transition, the mathematical decision rules, the protected Physical AI set, and the limits of interpreting record-level evidence as card-level taxonomy coverage.

**Keywords:** AI risk taxonomy; evidence synthesis; open-set classification; semantic mapping; Physical AI; responsible AI.

## 1 Introduction

A taxonomy can appear complete while concealing uncertainty in its source corpus, unit of analysis, or placement rules. RAI Risk Taxonomy 2.0 therefore separates four quantities that must not be conflated: the size of the integrated AI evidence space, the size of the AI-risk evidence view, the number of canonical L4 risk cards, and the number of L4 cards assigned to L3. The central result is a top-down evidence-to-taxonomy chain of 3,906,767 records, 82,971 AI-risk records, 1,726 candidate L4 rows, 1,725 exact-deduplicated L4 cards, and 1,725 L3-assigned cards.

The transitions do not all preserve the same statistical unit. The first two stages count documents or patent records. The third and fourth stages count canonical risk concepts. Consequently, the 82,971 evidence records are not a one-to-one parent table from which every L4 card was mechanically generated. They form an auditable evidence view supporting corpus coverage and risk interpretation, whereas the L4 registry integrates risk taxonomies, benchmarks, standards, and linked evidence through a source crosswalk.

## 2 Top-down dataset construction

### 2.1 Stage A: integrated AI evidence space

The top-level dataset contains 3,906,767 records from three source families: 1,749,159 papers, 125,372 policy reports, and 2,032,236 patents. Paper records originate from the local global Web of Science AI collections. Policy records combine the reviewed curated AI-policy master with the broader policy-materials collection. Patent records originate from the stage-2 AI patent master.

Records are standardized to a common schema and retained only when a non-empty title is available. Deduplication is performed within document type and analytical category. The ordered key hierarchy is patent application ID for patents, followed by DOI, source record ID, URL, and a normalized title–year–institution fallback. Cross-category overlap is intentionally retained because AI, AI infrastructure, Physical AI, and AI risk are analytical views rather than mutually exclusive partitions.

## 2.2 Stage B: integrated AI-risk evidence view

The AI-risk evidence view contains 82,971 records: 50,375 papers, 938 policy reports, and 31,658 patents. Its inclusion rule is source-governed rather than a single keyword query. Papers combine curated academic risk references, the clean and integrated AI-risk paper corpora, Physical AI safety benchmarks, and verified paper records from the top-200 evidence set. Policy reports combine curated risk-policy references with verified report records. Patents combine curated risk-patent references with verified patent records. Record-type filters are applied before the same title-validity and within-category deduplication rules used at Stage A.

This view represents 2.1% of the integrated AI evidence space. Because the source collections and analytical categories can overlap, this percentage is a descriptive ratio, not the estimated prevalence of risk research in all AI scholarship.

## 2.3 Stage C: canonical L4 registry

The unit changes from evidence record to risk concept. Source risk IDs are joined through the source crosswalk, semantically redundant cards are reviewed, and one canonical row is retained per L4 failure mode or harm mechanism. Permanent identifiers RAI4-0001 through RAI4-1726 are allocated independently of the risk label. The initial result is 1,726 candidate L4 rows. Automatic deduplication is restricted to pairs whose normalized label and mechanism-only definition are both exactly equal; semantic-similarity thresholds are not used. This merges RAI4-1606 into RAI4-1083, retains both source references, and yields 1,725 canonical cards. The retired ID remains in the crosswalk. The registry includes the 182-card Physical AI taxonomy as a protected mapping set.

The ratio of L4 cards to AI-risk evidence records is 2.1%. It is reported only as a scale-change diagnostic: one document may support multiple risks, and one risk may be supported by multiple documents. It is not a document acceptance rate.

## 2.4 Stage D: L3 assignment policy

The final policy view assigns all 1,725 canonical L4 cards to one of 50 L3 nodes (100.0%). Following independent expert review, 83 cards (4.8%) retain a `decision_required` marker, displayed as `HOLD`. This marker is metadata on an assigned card, not an L1, L2, or L3 taxonomy node.

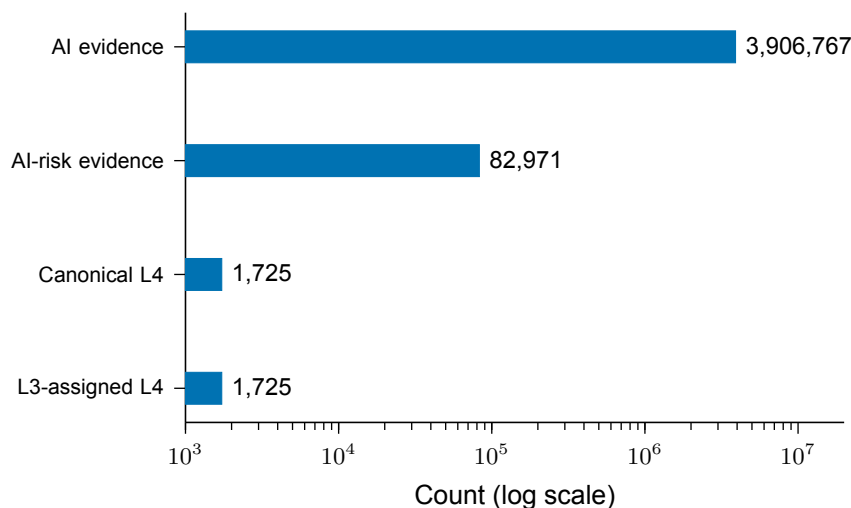


Figure 1: **Top-down evidence and taxonomy funnel.** Counts move from document/patent records to canonical risk concepts at the transition from AI-risk evidence to L4. The logarithmic axis preserves visibility across four orders of magnitude.

Table 1: Top-down stages, counts, and inclusion filters.

| Stage            | Count     | Unit                      | Principal filter                                                                                                                      |
|------------------|-----------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| AI evidence      | 3,906,767 | Document or patent record | AI source collections; valid title; standardized schema; within-type/category deduplication.                                          |
| AI-risk evidence | 82,971    | Document or patent record | Curated/integrated risk sources; record-type inclusion; valid title; within-type/category deduplication.                              |
| Canonical L4     | 1,725     | Risk card                 | Exact normalized label and core-definition match only; one retired ID mapped to its canonical card; all 182 Physical cards preserved. |
| L3 assigned      | 1,725     | Risk card                 | Protected Physical mapping or staged non-Physical placement; 83 decision-required cards remain assigned and marked HOLD.              |

### 3 Mathematical classification procedure

#### 3.1 Protected Physical AI mapping

The 182 L4-to-L3 mappings inherited from the Physical AI taxonomy are treated as the locked gold set. Later stages cannot change these paths. The remaining 1,544 cards enter the hierarchy-blind classification procedure. Legacy global L0–L3 labels, source families, and source-ID prefixes remain reference metadata and are excluded from prediction features.

#### 3.2 Stage 1: strict open-set gate

The input for each non-Physical card is the L4 label, a mechanism-only definition with hierarchy restatements removed, and the evidence title. Let  $s_{sem}$  be the top-1 semantic score,  $s_{comp}$  the top-1 composite score,  $m_{sem}$  and  $m_{comp}$  their respective top-1 margins, and  $v_{anchor}$  the number of anchor views selecting the same L3. A candidate is retained only when all seven conditions hold:

1. exactly one L3 is rule-eligible;
2. the rule winner, semantic top-1, and composite top-1 are identical;
3.  $s_{sem} \geq 0.55$  and  $s_{comp} \geq 0.54$ ;
4.  $m_{sem} \geq 0.035$  and  $m_{comp} \geq 0.035$ ;
5.  $v_{anchor} \geq 2$  across three anchor views;
6. no gap sentinel, multi-mechanism flag, or Physical-boundary flag is present; and
7. two independent expert agents approve the same exact L3.

This gate yields 37 new expert-consensus proposals. Together with the 182 Physical locks, Stage 1 assigns 219 cards and withholds 1,507.

#### 3.3 Stage 2: suitability-ranked relaxation

Stage 2 preserves all 219 protected assignments and computes

$$S_2 = 0.55q_{sem} + 0.20q_{sm} + 0.10q_{cm} + 0.10q_{vote} + 0.05q_{rule} - p_{gap}, \quad (1)$$

where

$$q_{sem} = \text{clip} \left( \frac{s_{sem} - 0.45}{0.25}, 0, 1 \right), \quad (2)$$

$$q_{sm} = \text{clip} \left( \frac{m_{sem}}{0.05}, 0, 1 \right), \quad (3)$$

$$q_{cm} = \text{clip} \left( \frac{m_{comp}}{0.05}, 0, 1 \right), \quad (4)$$

$$q_{vote} = v_{anchor}/3, \quad (5)$$

$$q_{rule} = \mathbf{1}[\text{at least one eligible L3}], \quad (6)$$

$$p_{gap} = 0.05 \mathbf{1}[\text{gap sentinel present}]. \quad (7)$$

$S_2$  is a deterministic ranking score rather than a calibrated probability. Stage 2 first preserves 55 hard holds. It then orders the remaining holds by ascending  $S_2$ , with permanent RAI4 ID as the tie breaker, and reserves the lowest 118. Thus  $55 + 118 = 173$  cards remain unresolved and 1,553 cards are assigned (90.0%).

### 3.4 Stage 3: forced-path test and final policy

Stage 3 preserves the 1,553 Stage 2 paths and assigns only the 173 unresolved cards to the hierarchy-blind top-1 L3 already stored in the score artifact. This produces 1,726 operational paths but does not validate taxonomy fit. Each new path retains its original hold reason, score, review priority, and `human_approved=false` marker.

The final policy marks five difficult-case families for decision review: 23 Physical-scope reviews, 17 expert rejections, 7 expert disagreements, 6 multi-mechanism cases, and 2 low-absolute-fit cases. Their forced L3 path remains populated and `decision_required=true`. The remaining 118 quota holds also retain their forced paths and review metadata.

### 3.5 Independent expert card review

Two reviewers independently inspected all 1,725 cards for agreement among the L4 label, mechanism definition, evidence, and assigned L3. Reviewer A proposed 30 amendments and Reviewer B proposed 14. Automatic amendment required both reviewers to propose the same label and L3 at high confidence. One card met this rule: **RAI4-0888** changed from “Privacy concerns” to “Anthropomorphic trust-induced privacy disclosure”, while retaining **RAI3-G-INT-10**. Its original label remains in provenance metadata. Forty-two cards had a proposal from only one reviewer or incompatible proposals; their labels and paths were not changed. Thirteen were already marked **HOLD** and 29 newly received the marker. The consensus amendment cleared the prior hold on **RAI4-0888**, giving  $55 - 1 + 29 = 83$  final **HOLD** cards. No protected Physical AI card was changed.

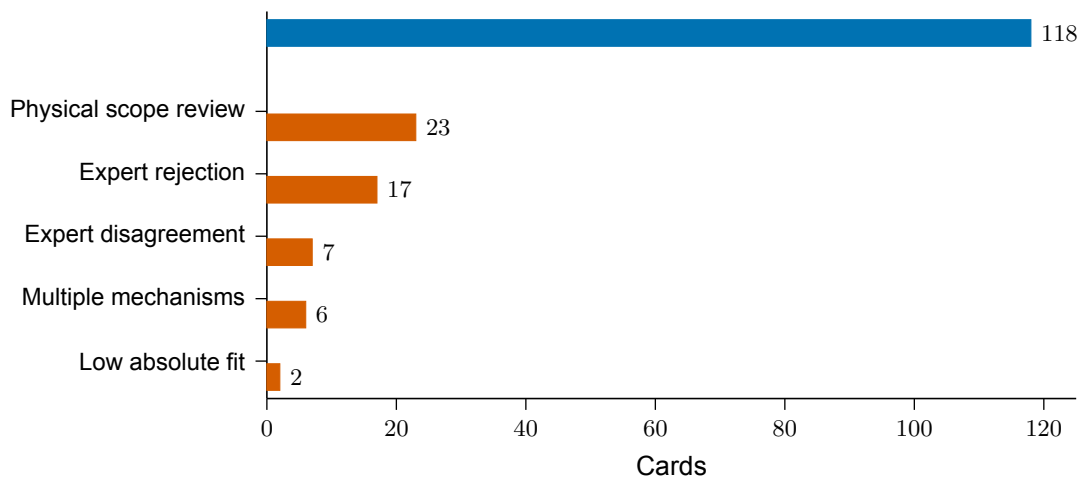


Figure 2: **Disposition of the 173 Stage 2 holds.** The 55 difficult cases (vermillion) are force-assigned and marked HOLD; 118 suitability-reserve cards (blue) are also force-assigned.

## 4 Results

### 4.1 Evidence composition

Papers account for 44.8% of the integrated AI evidence space, policy reports for 3.2%, and patents for 52.0%. Within the AI-risk evidence view, the corresponding shares are 60.7%, 1.1%, and 38.2%. These shares use one decimal because their integer component is at least one.

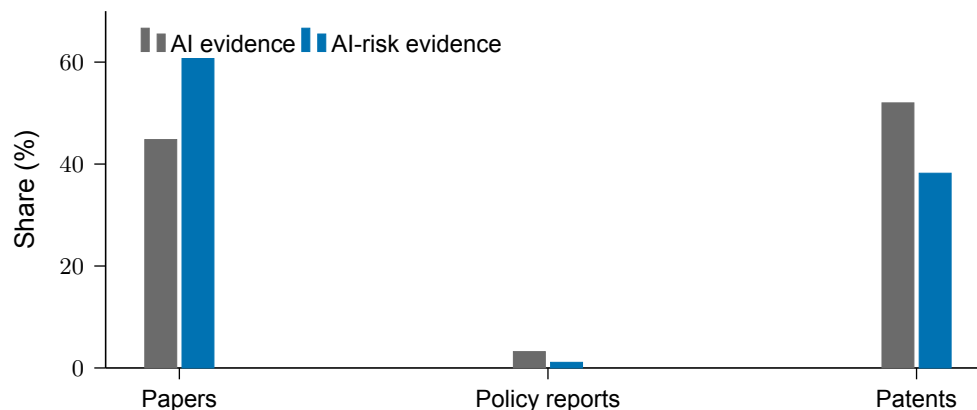


Figure 3: **Source-family composition.** Percentages compare document-type composition within the integrated AI evidence space and the AI-risk evidence view.

### 4.2 Final hierarchy coverage

The final policy assigns all 1,725 canonical cards (100.0%); 83 assigned cards (4.8%) carry the **HOLD** marker. Of all cards, 1,191 are General, 352 are Agentic AI, and 182 are Physical AI. Their shares are 69.0%, 20.4%, and 10.6%, respectively. All 50 L3 nodes contain at least one card. Anthropomorphism is the largest L3 with 254 cards (14.7%); the five largest L3 nodes jointly contain 42.0%.

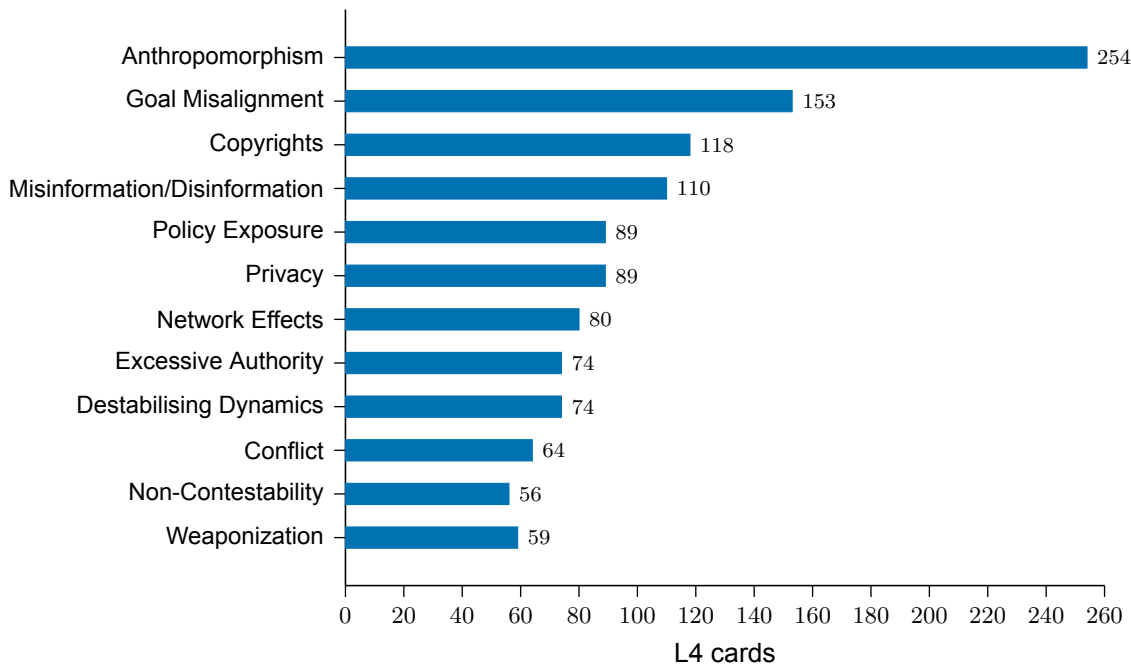


Figure 4: **Largest L3 nodes in the final policy view.** The figure shows the twelve largest L3 nodes. Counts include the 83 assigned cards marked **HOLD**.

## 5 Rounding and reporting policy

All counts are reported as integers. Percentages below 1 are rounded at the third decimal place and displayed to two decimal places; percentages with an integer component of at least 1 are rounded and displayed to one decimal place. For example, the L4 registry represents 0.04% of the integrated AI evidence count, the AI-risk view represents 2.1%, final L3 coverage is 100.0%, and the decision-required share is 4.8%. Exact floating-point ratios remain available in the machine-readable statistics artifact.

## 6 Validation and limitations

The 182 Physical mappings and 37 Stage 1 expert-consensus mappings remain unchanged in later stages. Repeated Stage 2 and Stage 3 builds produce byte-identical placement hashes. The Stage 2 validator reports 13 PASS, 0 FAIL, and 3 WARN; the Stage 3 validator reports 14 PASS, 0 FAIL, and 4 WARN. All 18 repository tests pass at the time of this report revision.

Three limitations directly affect interpretation. First, cross-category overlap is retained in the integrated evidence space, so category counts are analytical views rather than mutually exclusive samples. Second, the transition from 82,971 evidence records to 1,725 canonical L4 cards changes the unit of analysis and is not a mechanical screening step. Third, 1,214 of the 1,334 new Stage 2 placements lack Stage 1 rule eligibility, 247 override a taxonomy-gap sentinel, and the remaining 118 Stage 3 forced candidates are not human-approved ground truth.

## 7 Review priorities

The 83 assigned cards marked **HOLD** should be reviewed first, beginning with the 42 cards lacking two-reviewer consensus and the original hard-hold cases. The 118 quota candidates should then be reviewed in ascending  $S_2$  order. Finally, the 247 gap overrides and 1,214 rule-free placements should be audited through L3-stratified samples. Human review should determine whether each failure is best addressed by expanding an L3 definition, introducing a new L3, or allowing primary and secondary labels.

## Data availability

The exact top-down counts, filters, placements, and public card bundle are available in the repository:

- `reports/statistics/top_down_evidence_funnel.json`
- `data/experiments/stage3-forced-policy-v2/placements.json`
- `public/data/releases/v2.2.0/cards.json`
- `reports/data_quality/l4_deduplication_v2.1/retired_to_canonical.json`
- `reports/expert_review/v2.1/consensus_result.json`

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